

PROGRAMMING TIMERS IN C

C Programming of Timers as Counters (cont')

Example 9-27

Assume that a 1-Hz external clock is being fed into pin T0 (P3.4). Write a C program for counter 0 in mode 1 (16-bit) to count the pulses and display the state of the TH0 and TL0 registers on P2 and P1, respectively.

Solution:

```
#include <reg51.h>
void main(void) {
    T0=1;
    TMOD=0x05;
    TL0=0;
    TH0=0;
    while (1) {
        do {
            TR0=1;
            P1=TL0;
            P2=TH0;
        }
        while (TF0==0);
        TR0=0;
        TF0=0;
    }
}
```